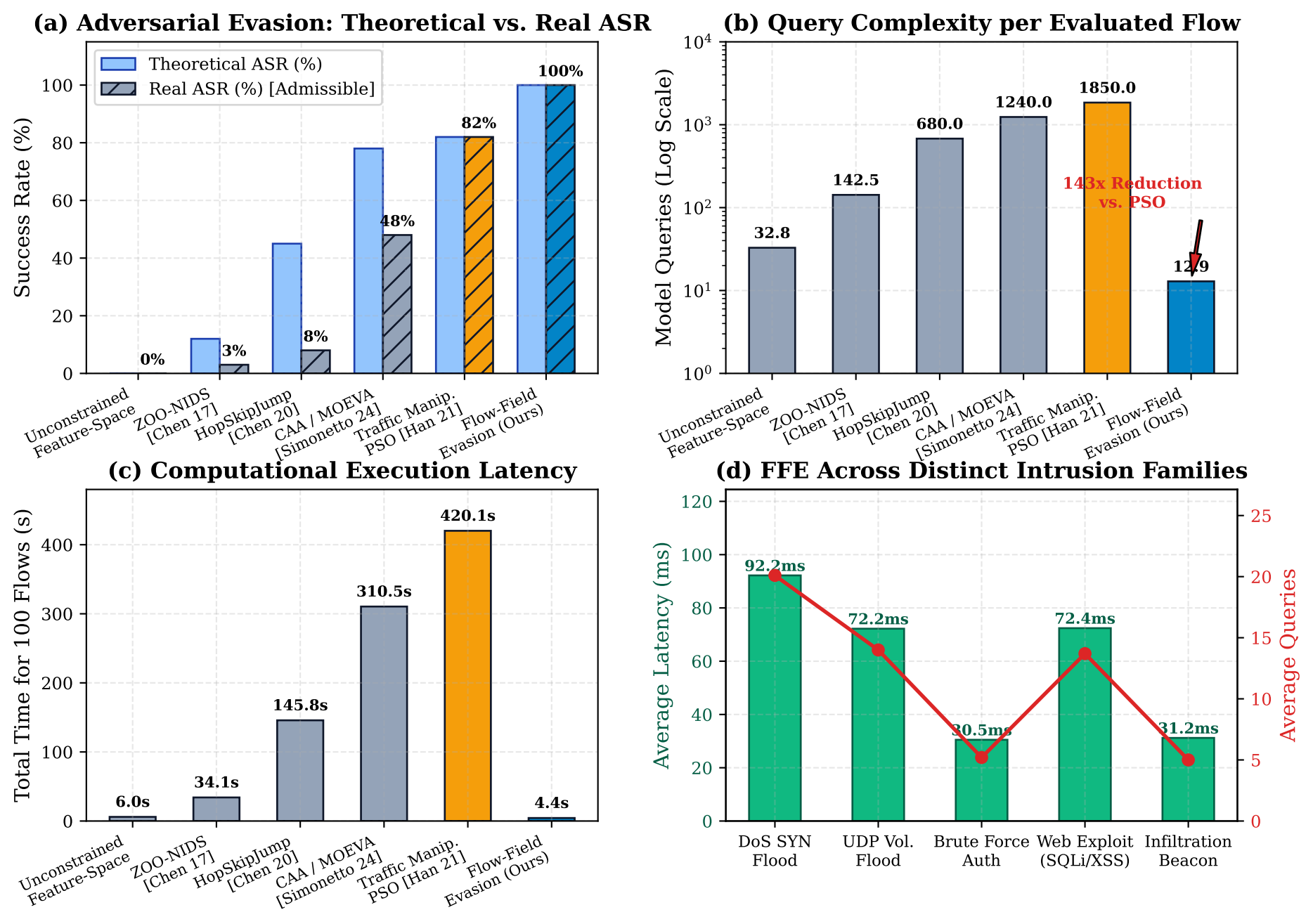
(a) Adversarial Evasion: Theoretical vs. Real ASR
Theoretical ASR (%)
Real ASR (%) [Admissible]
Success Rate (%)
0%
3%
8%
48%
82%
100%
(b) Query Complexity per Evaluated Flow
Model Queries (Log Scale)
32.8
142.5
680.0
1240.0
1850.0
143x Reduction vs. PSO
12.9
(c) Computational Execution Latency
Total Time for 100 Flows (s)
6.0s
34.1s
145.8s
310.5s
420.1s
4.4s
(d) FFE Across Distinct Intrusion Families
Average Latency (ms)
Average Queries
92.2ms
72.2ms
30.5ms
72.4ms
31.2ms
DoS SYN Flood
UDP Vol. Flood
Brute Force Auth
Web Exploit (SQLi/XSS)
Infiltration Beacon
Unconstrained Feature-Space
ZOO-NIDS [Chen 17]
HopSkipJump [Chen 20]
CAA / MOEVA [Simonetto 24]
Traffic Manip. PSO [Han 21]
Flow-Field Evasion (Ours)